DSCI 560
Lecture 13: Internet of Things

Data Science Practicum

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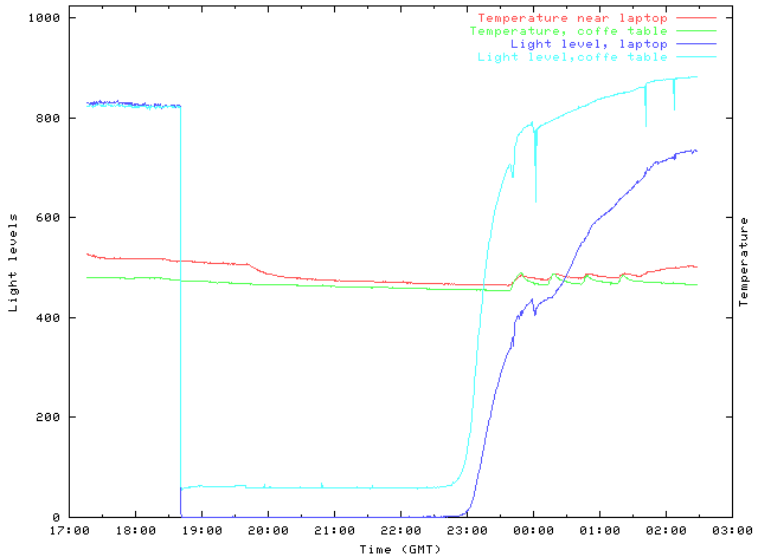
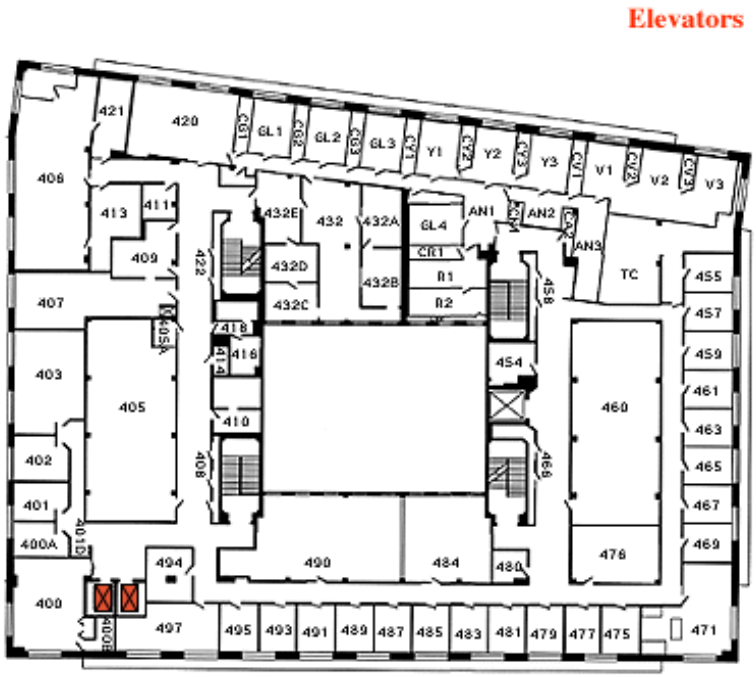
The Vision for Internet of Things

- Push out computers into the real world
- Billions of sensors and actuators
- Zero configuration
- Merging Cyberspace & Physical space

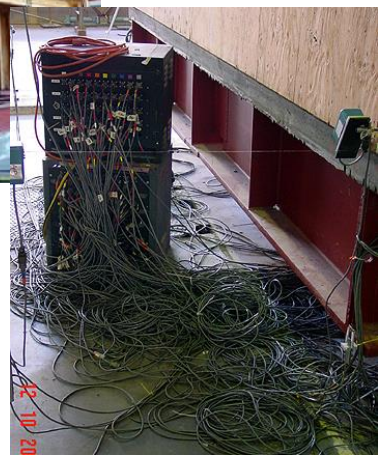
Energy Monitoring/Mgmt System



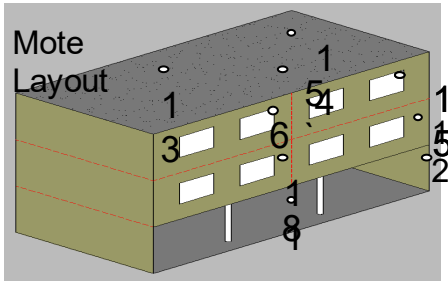


- 50 nodes on 4th floor
- 5 level ad hoc net
- 30 sec sampling
- 250K samples to database over 6 weeks

Structural performance due to multi-directional ground motions (Glaser & CalTech)

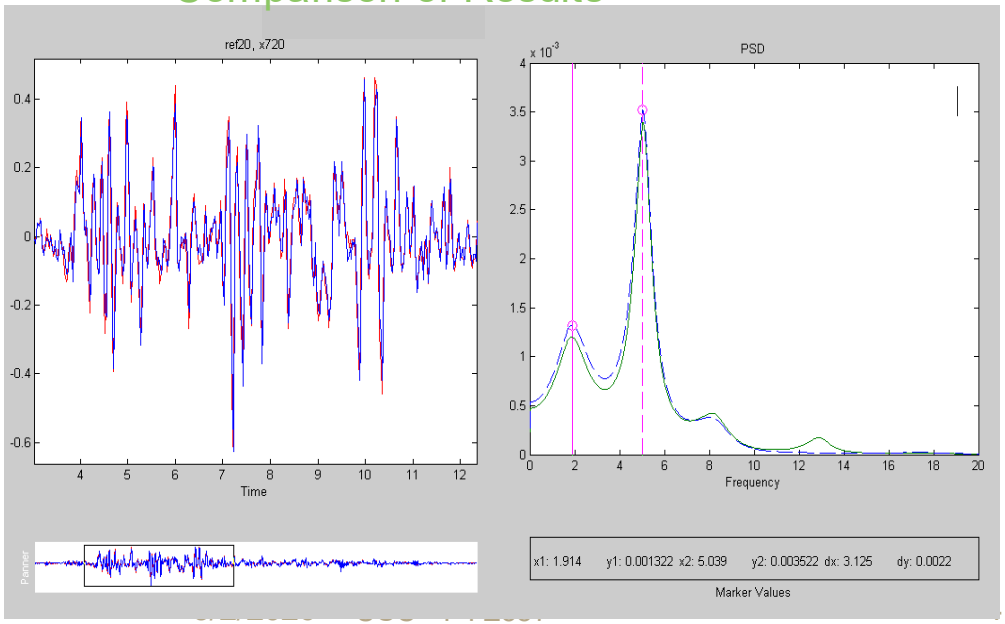


Wiring for traditional structural instrumentation + truckload of equipment



Mote infrastructure

Comparison of Results



Network Algorithms

- Adaptive Routing
- Network Query Processing
- Distributed Data Aggregation
- Localization/Time Synchronization
- Power Management
- Compression
- Security

Node Requirements

- Low Power, Low Power, Low Power…
- Support Multi-hop Wireless Communication
- Self Configuring and Small Physical Size
- Programmable Processor
- Integrated Sensors
- Meets Research Goals
  - Operating system exploration
  - Enables exploration of algorithm space
  - Instrumentation
  - Network architecture exploration

System Architecture

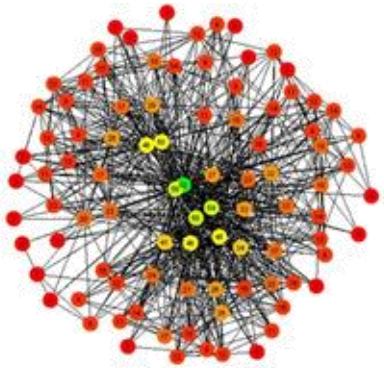
- Node Architecture
  - Hardware architecture: how different hardware components interact with each other
  - Software architecture: how software pieces are put together for various functionalities
- Network architecture
  - How nodes interact through different topologies
  - How different protocols coordinate such interaction

The Nodes

- Alternative Names
  - Wireless Sensor Network
  - Cyber-Physical Systems
  - Internet of Everything
- Sensing the Environment
  - Temperature, pressure, flow rate, etc
  - Calibration
- Intelligent Data Processing
  - Data processing/forming
  - Detect problems locally or with neighbors
  - Proactively report exceptions
- Collaborative Networking
  - Does not require special infrastructure
  - Sensor nodes relay data for other nodes

Composition of Node

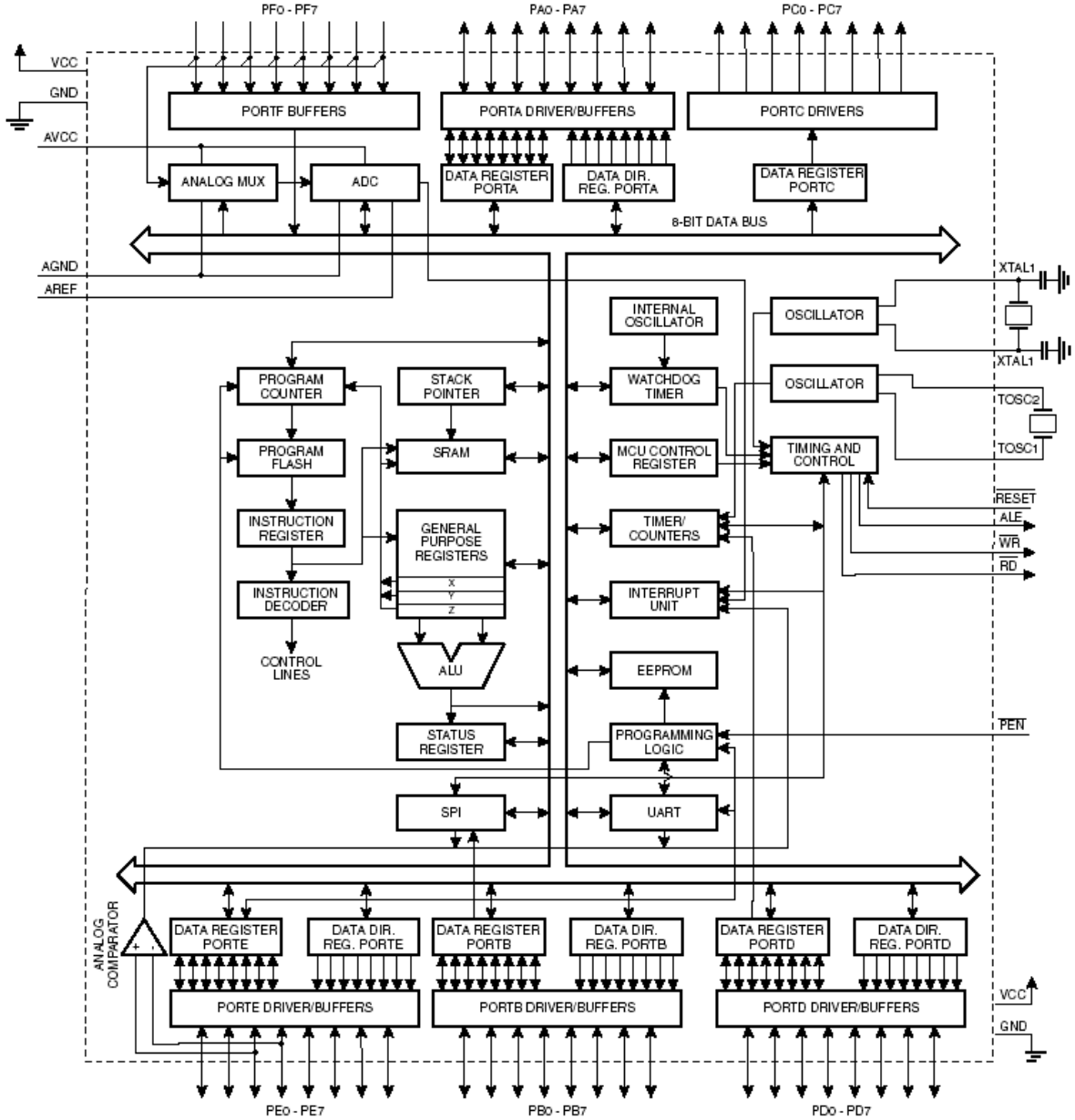
- Sensor
  - Detect/Measure Physical Phenomenon
  - Variety of Responses
  - Different Types of Sensors
- Network
  - Communications between the Nodes
  - Multiple Wired/Wireless Nodes
  - Data Distribution
- Intelligence
  - General Purpose Processor
  - Application Specific Logic
  - Software



The Processor

- Intelligence in the Node
- Important Questions
  - What's inside a microcontroller today?
  - What peripheral support?
  - How do you program one?
  - How much memory?
  - Programming platforms?

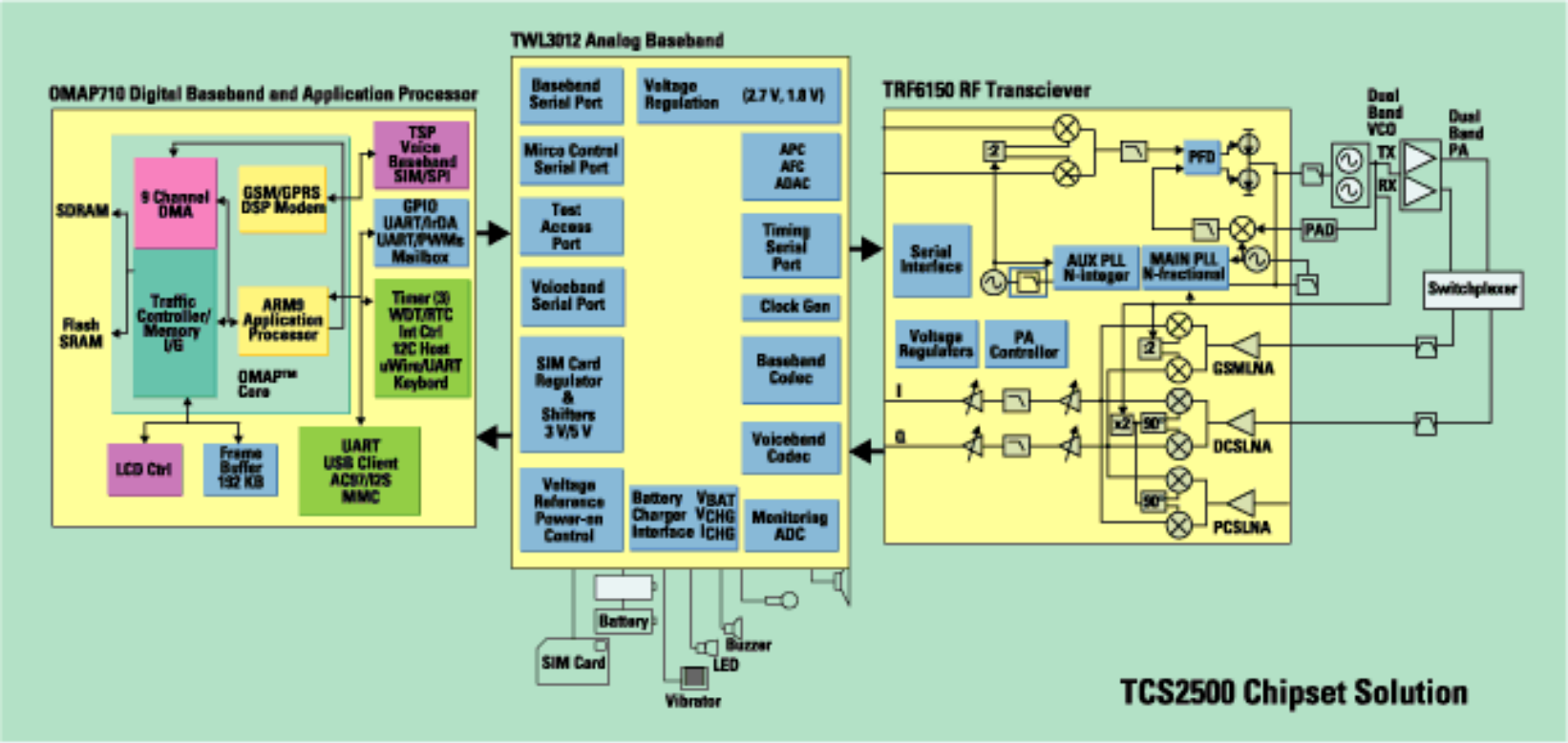
Figure 1. The ATmega103(L) Block Diagram



The Communication

- Major RF Devices
  - Cordless Phones Digital/Analog
    - Single Channel
  - Cellular Phones
    - Multi-channel, Base station controlled
  - 802.11
    - “wireless Ethernet”
  - Bluetooth
    - Emerging, low-power frequency hopping

Mobile Phone: An Ultimate IoT

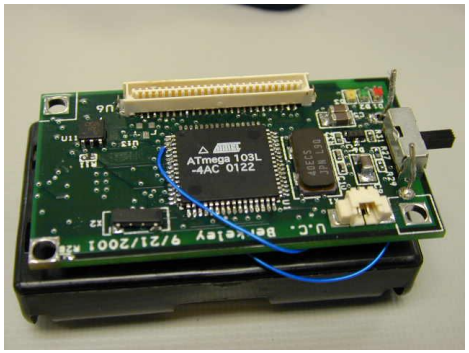


Processor: ARM9, 120Mhz + DSP

Communication: 270.833 kbps RF

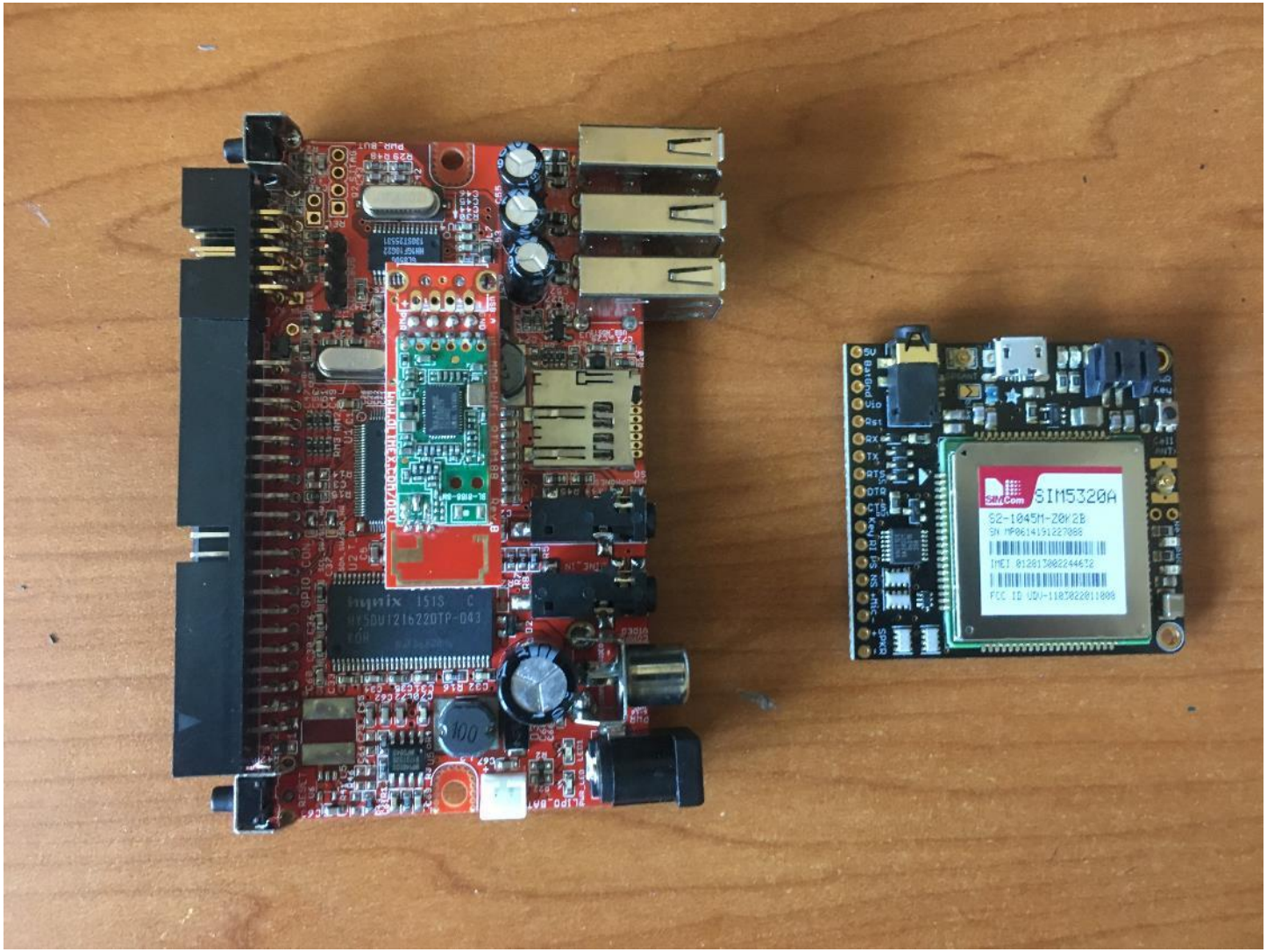
Early Research Wireless IoT

- Atmel ATMEGA103
  - 4 Mhz 8-bit CPU
  - 128KB Instruction Memory
  - 4KB RAM
  - 5.5mA active, 1.6mA idle, <1uA powered down
- 4 Mbit flash (`AT45DB041B`)
  - SPI interface
  - 1-4 uj/bit r/w
- RFM TR1000 radio
  - 50 kb/s – ASK
  - Communication focused hardware acceleration
- Network programmable
- 51-pin expansion connector
  - Analog compare + interrupts
- GCC based tool/programming chain



Cost-effective power source
2xAA form factor

Example: iMX233-OLinuXino-MINI + FONA 3G



i.mx233 Olimex-Mini

- Embedded Linux
  - Custom drivers installed (audio, I2C, SPI)
  - Compiled and functioning
- Audio
  - Able to play stereo audio
  - Able to record stereo audio using ARECORD into WAV
  - Need to compress audio before transmission
- Communication
  - Driver for 3G UART needs to be written
  - Need to test and evaluate

Example: Nokia 8110 4G Phone

- Installed Custom OS
  - (GerdaOS – derivative of KaiOS)
  - Rooted
- Telnet
  - Telnet server installed
  - Successful telnet to phone's root

```
root@Nokia 8110 4G:/ #
```

Nokia 8110 4G Power Control

- Processor Speed
  - Root shell to reduce the operating frequency of the processor from 800 MHz down to 200 MHz (minimum functional operating frequency)

```
[t /sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq
200000
root@Nokia 8110 4G:/ #
```

- Core Activation/Deactivation
  - Implemented a method via the root shell to turn off one of the CPU's cores
  - Below are screenshots showing the second core (cpu1) becoming offline

```
[root@Nokia 8110 4G:/sys/devices/system/cpu # cat online
0-1
[root@Nokia 8110 4G:/sys/devices/system/cpu # cat offline
```

```
[root@Nokia 8110 4G:/sys/devices/system/cpu # cat online
0
[root@Nokia 8110 4G:/sys/devices/system/cpu # cat offline
1
```

Nokia 8110 4G Remote Access

- No Static IP Address for Phone
  - Most phone plans do not give static IP
  - Need a work around to access phone remotely
  - Use of VPN Tunnel will enable this access
- Enabling VPN to Phone
  - Successfully initialized a VPN connection between the phone and a remote node
  - Currently searching for a method to disable/bypass a firewall on the phone that is blocking VPN outgoing messages

Constraints

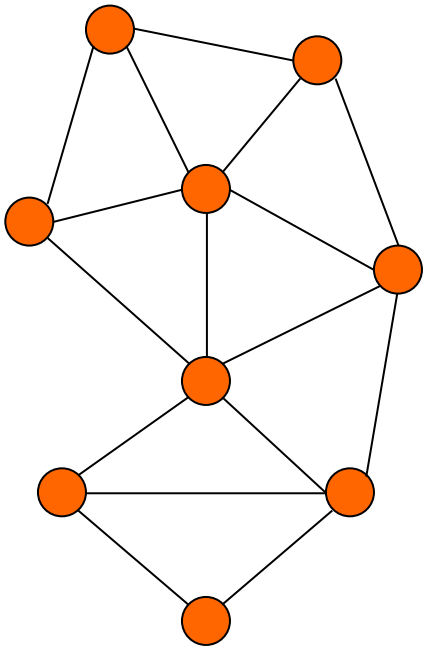
- After Building Applications
  - System is highly memory constrained
  - Communication bandwidth is limited by CPU overhead at key times.
  - Communication has bursty phases.
- Where did the Energy/Time go?
  - 50% of CPU used when searching for packets
  - With 1 packet per second, >90% of energy goes to RX!

The Most Important Factor

- POWER, POWER, POWER
- A Decade of Research: 90s-2000
  - Borrowed concepts from Distributed Systems
  - ALL with targeted focus on low power use
- Minimize Power for the Same Result?!?
  - Communication: the biggest power consumer
  - Then processor
  - Then peripherals/sensors

Key Function 1: Localization

- GPS Localization
  - Use the GPS coordinates
    - Additional GPS units
    - High Cost in Area and Energy
    - Not always available
- GPS-less Localization
  - Find Relative Angle
    - Where the angular vectors meet
  - Find Distances and estimate coordinates
    - Use Radio and/or Ultrasound
    - Additional localization algorithm
    - Low Cost in Area and Energy
    - Available within range

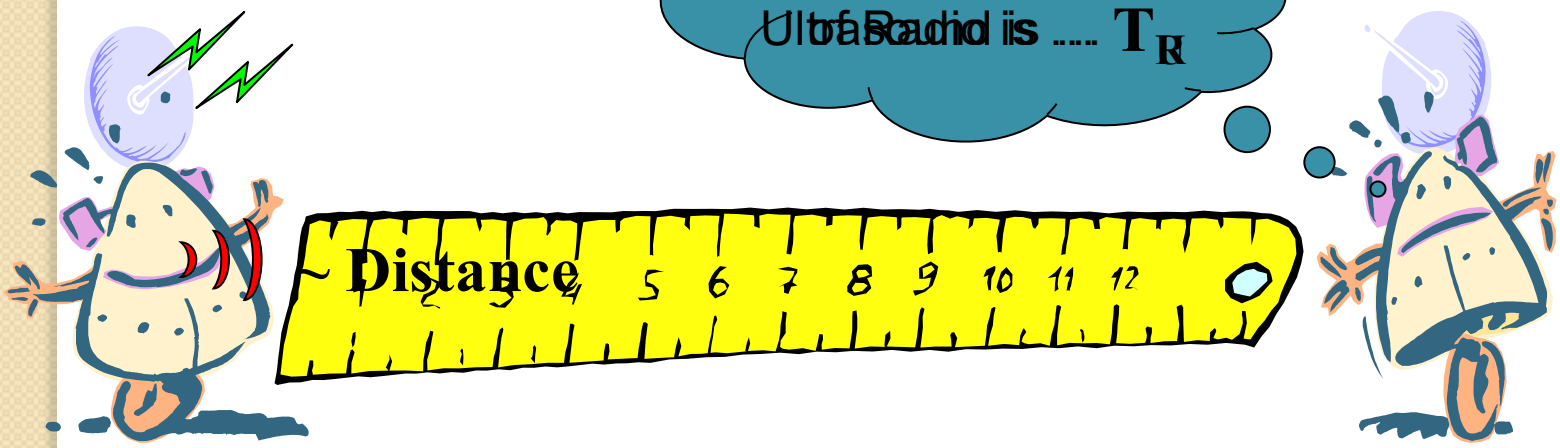


TDoA: Distance Calculation

Radio Transmitter

Time of Arrival of Radio is ..... T_R

Radio Receiver



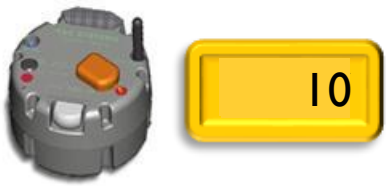
Ultrasound Transmitter

Ultrasound Receiver

$$\frac{\text{Distance}}{\text{Speed of Sound}}$$

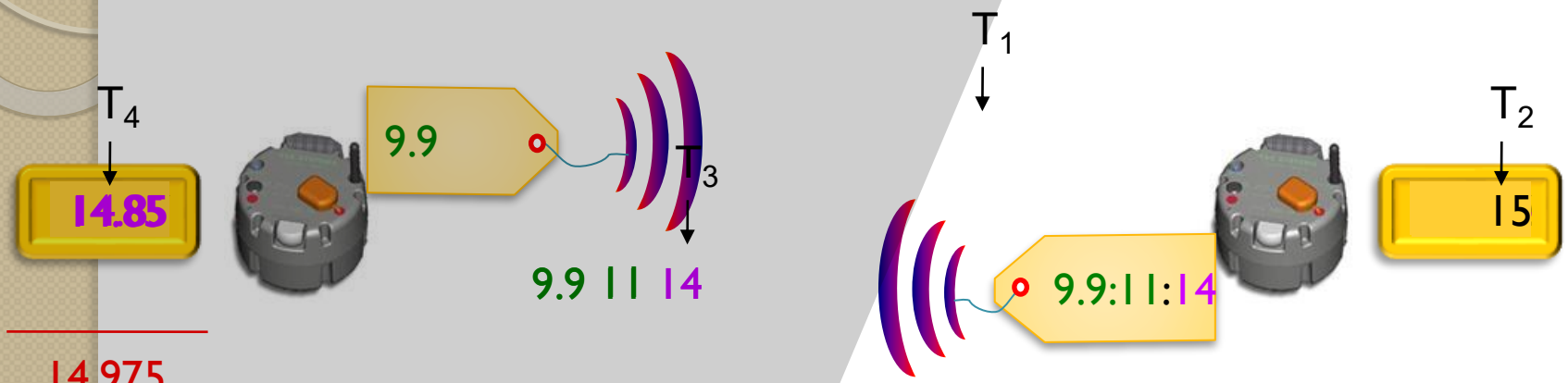
$$- \qquad = \text{Time Difference of Arrival}$$

Key Function 2: Time Synchronization



Drift is highly dependent on the environment (i.e. Temp)
Difference in drift can accumulate over Time

Time Synchronization



14.975

- $T_2 = T_1 + \text{Propagation Delay} + \text{Timer Offset}_1$
- $T_4 = T_3 + \text{Propagation Delay} - \text{Timer Offset}_2$
- Propagation Delay ≈ ((11 - 9.9) + (14.85 - 14)) / 2 = 0.975
- Timer Offset ≈ ((11 - 9.9) - (14.85 - 14)) 0.125 125

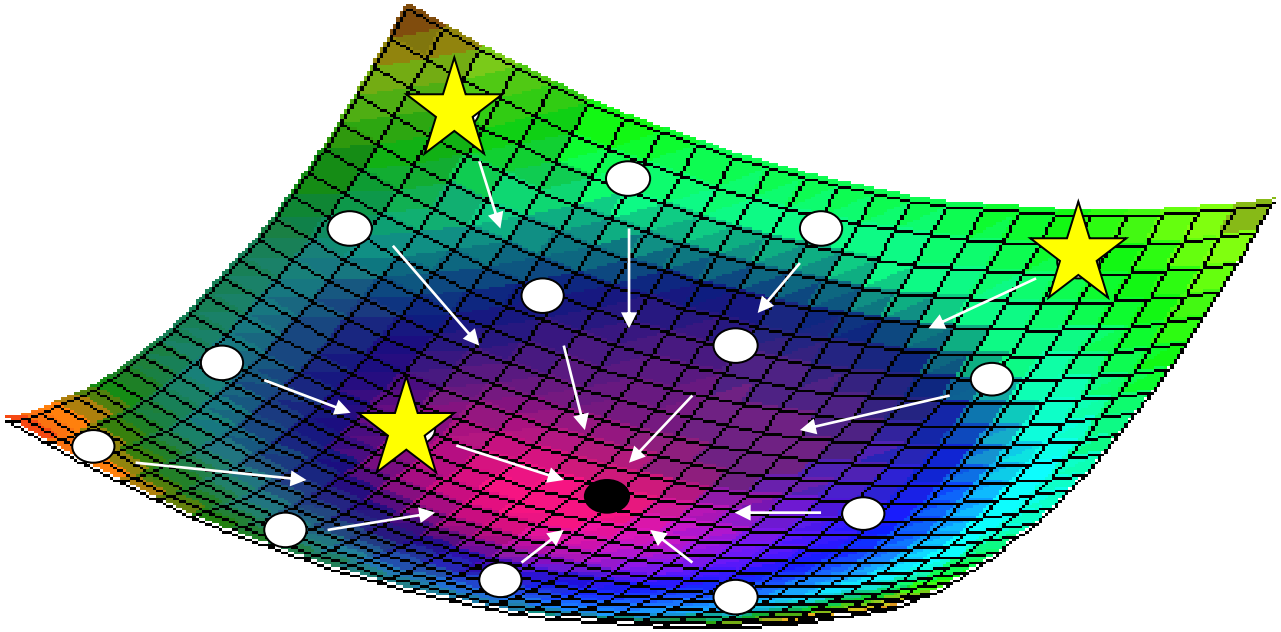
Not exactly equal due to frequency drift of the clocks during the flight times of messages

May solve the problem, BUT time synchronizations COST resources!!

For resource constrained systems like wireless sensor network, fewer time synch is better.

Key Function 3: Data Aggregation

- Example: Virtual Gradient based
  - Flow of water from mountain to valley
  - Gradient is setup from network to Sink



Summary

- Components of IoT
  - Processor
  - Communication
  - Software
- The Main Constraint
  - Low Power Consumption
  - Comm then Processor then Peripheral/Sensor
- Important Services
  - Localization
  - Time Synchronization
  - Data Aggregation

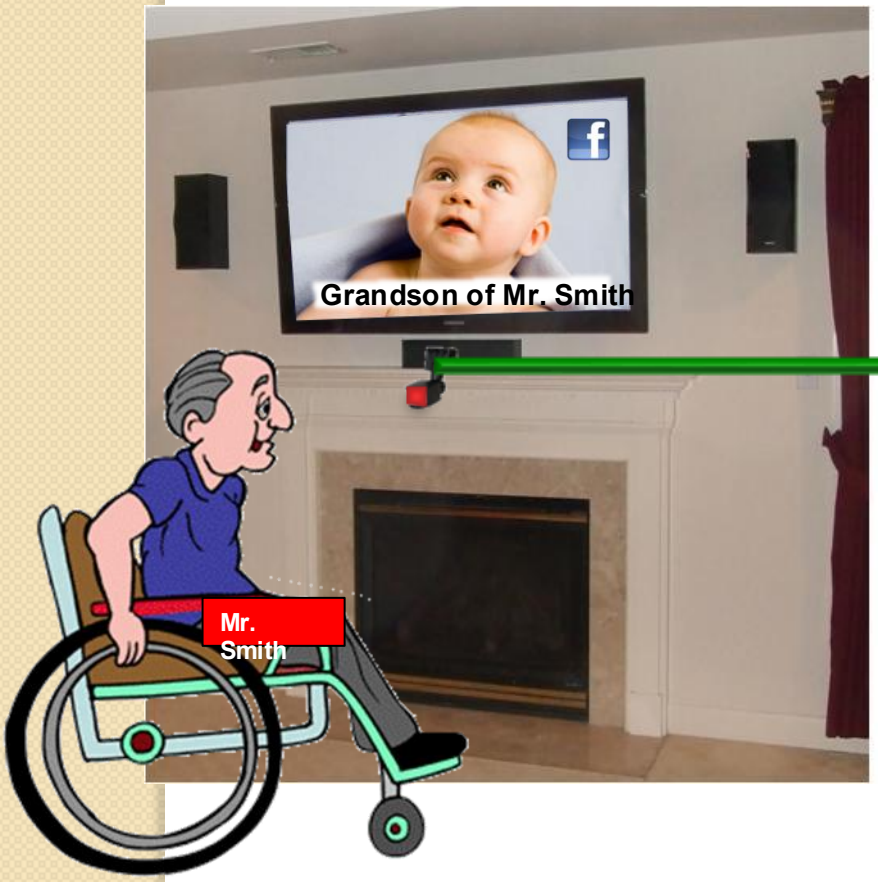
Example: Physical Space versus Cyberspace

Wide Range of Privacy Preserving Sensors and Natural User Interface

Intelligent and Adaptive User Profilers and Translation System

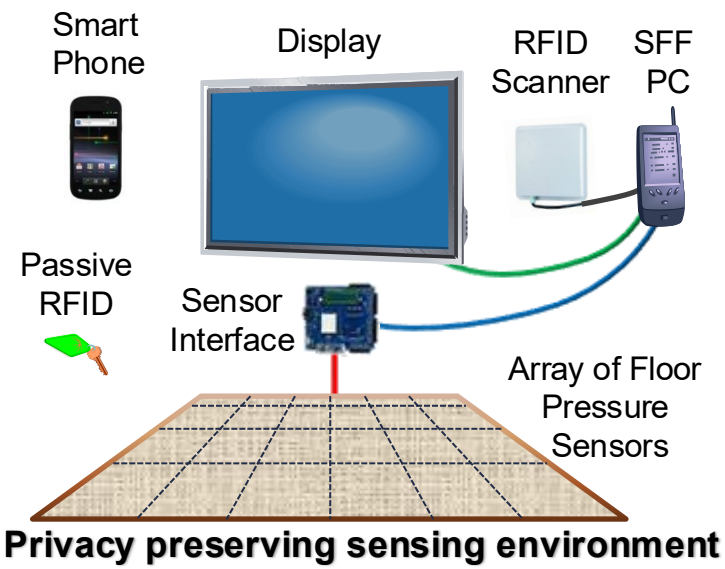
Seamless Merging of Cyber-Physical Space

- Allow people in physical space to receive cyberspace services without any effort
- Allow cyberspace members to interact with persons in physical space as other cyberspace members



Physical Space Interface

- Sensing Physical Space
  - Video cameras (big privacy concerns!)
  - Carpet sensors (position without ID)
  - Passive RFID (ID without position)
  - Options for degree of exposures
  - Anonymous user possible



- Natural User Interface
  - Use of pervasive appliances
  - User Input
    - Indirect input based on floor pressure sensor and sensor network data
    - Indirect inferential input based on behaviors (i.e. stopping to look or walk-by)
    - Direct input with simple button(s) or proximity sensors
    - Direct input using Smart cell phones or
  - Output
    - Wall mounted HD Televisions (display)
    - Stereo speakers

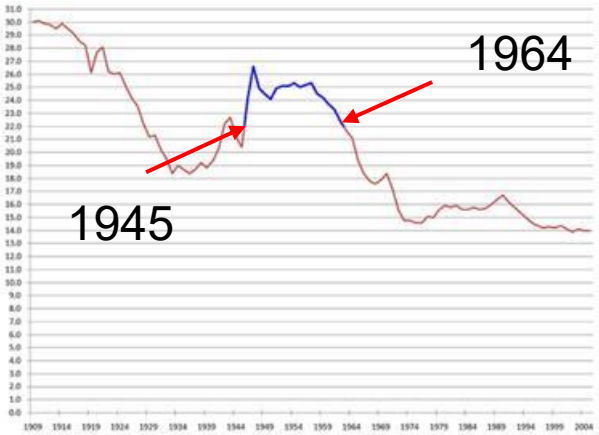
Cyberspace Interface

- Monitoring Cyberspace
  - Network traffic monitoring
  - Pattern matching

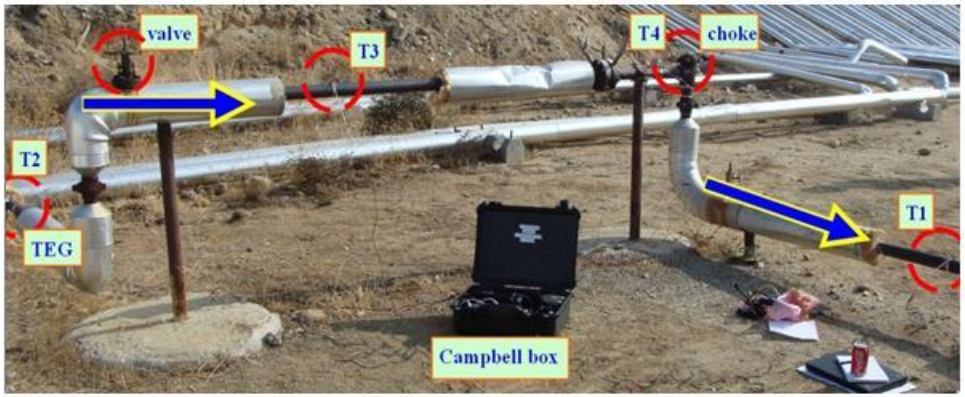
- Preservation of Privacy
  - Degree of privacy
  - Dynamically adjusting the output using sensor data
- Translation Systems
  - Language translation
  - Domain friendly conversion
- Automatic User Profile Builder
  - Inferential data from indirect input
  - Building accurate profiler using history

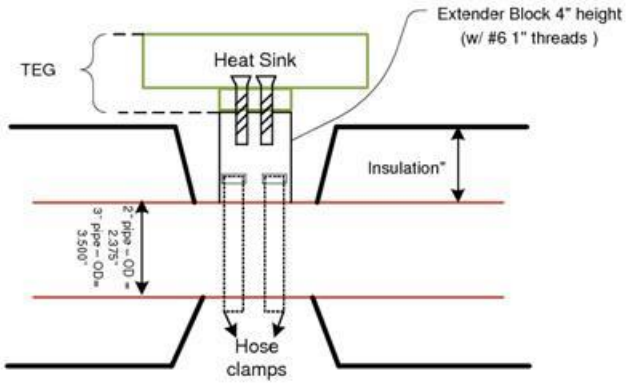
Cyber-Physical Assisted Living (CPAL)

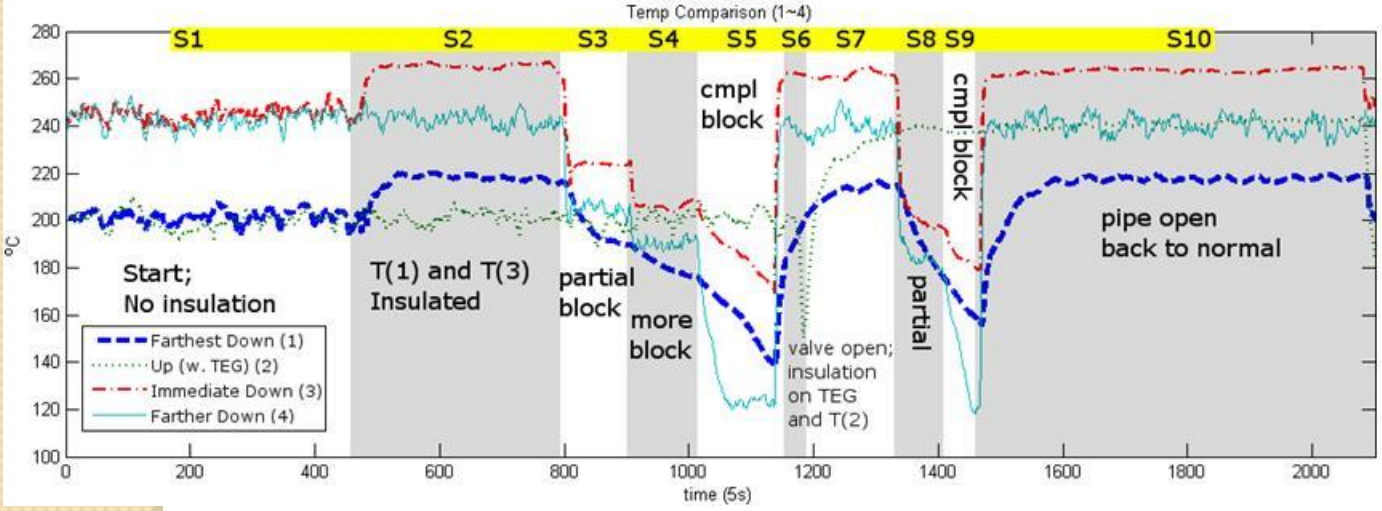
- Increase in Elderly Population
  - First of baby boomers retiring 2010
  - One of the largest generation
  - Declining health and loneliness
- Related Research
  - Ubiquitous Sensing
    - Mostly on health monitoring
  - Social well-being
    - No practical solution
    - physical visits and counseling
  - No research establishing the interface



Example: Steam-powered IoT Network

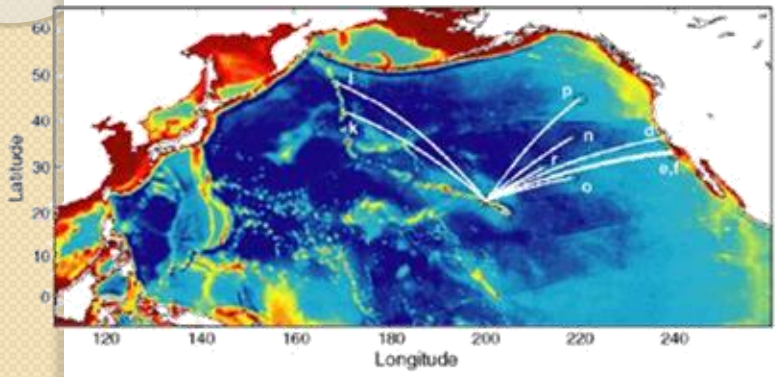


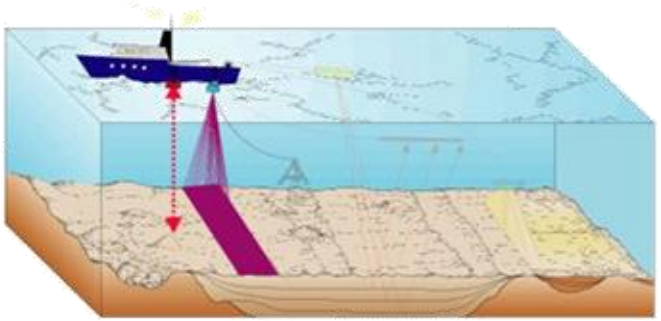




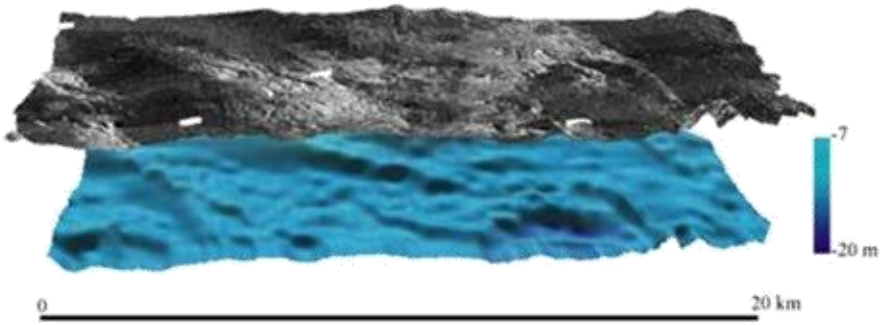
Underwater Sensor Network

- Example: Underwater Acoustic Sensing



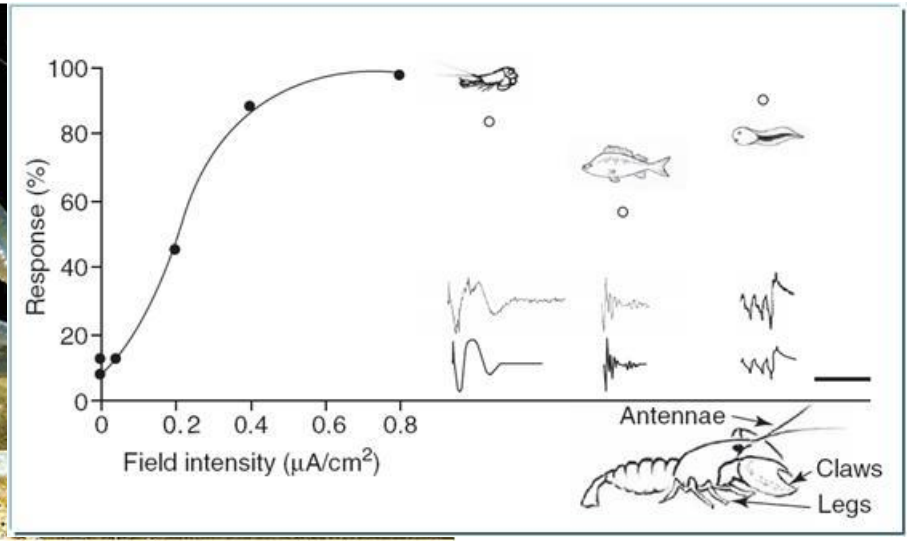


Micro-ocean acoustic tomography



Acoustic beamforming

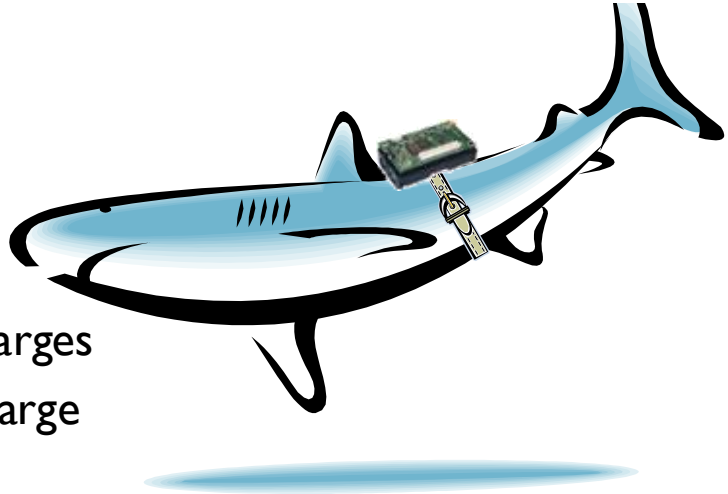
Australian Yabbies



- Ability to sense very low electrical field from muscle movement
  - Sharks also have this ability to find their prey using the same sense
  - Worth studying the physiology of Australian Yabbies (or Crayfish)
  - Patulio and Macmillan, "Crayfish respond to electrical fields," Current Biology, vol 17-3.
- Directly send electrical signal in seawater at high frequency
  - Seawater has higher electrical conductance than distilled water
  - Use seawater like a conductor – experiments show promising results
  - Preliminary experiments indicate μA signals can traverse meters of seawater without significant signal degradation (needs more experiment)

Unconventional Sensors

- Biological/Bioinspired Research
  - Lemon shark can detect tuna oil at 40 ppb
    - About 10 drops in an average-sized home swimming pool
    - Other types of sharks can detect at 0.1 ppb
  - Attaching sensors to marine animals (i.e. sharks)
  - Energy harvesting of water current
- Other Physical Qualities of Water
  - Localization based on ion concentration
  - Seawater working like capacitors
    - Preliminary experiments show lingering charges
    - Dissipates in duration depending on the charge
    - Leave pockets of ionized seawater
    - Sense the pockets to localize or communicate



Real World Deployment

- Field IoT Devices
  - No power plugs in Most Outdoors/Water
  - Need Continuous Monitoring
  - Power Problems
    - 10 years only for certain applications
    - Short Life in Practice (1-2 years typical)
- Goals
  - Net Zero Power
    - Lower power operation
    - Smart Processing at the Leaf Nodes (AI ?!?)
    - Remove the Batteries/Energy Harvesting
  - Automated Configuration and Calibration
  - Maintenance Free

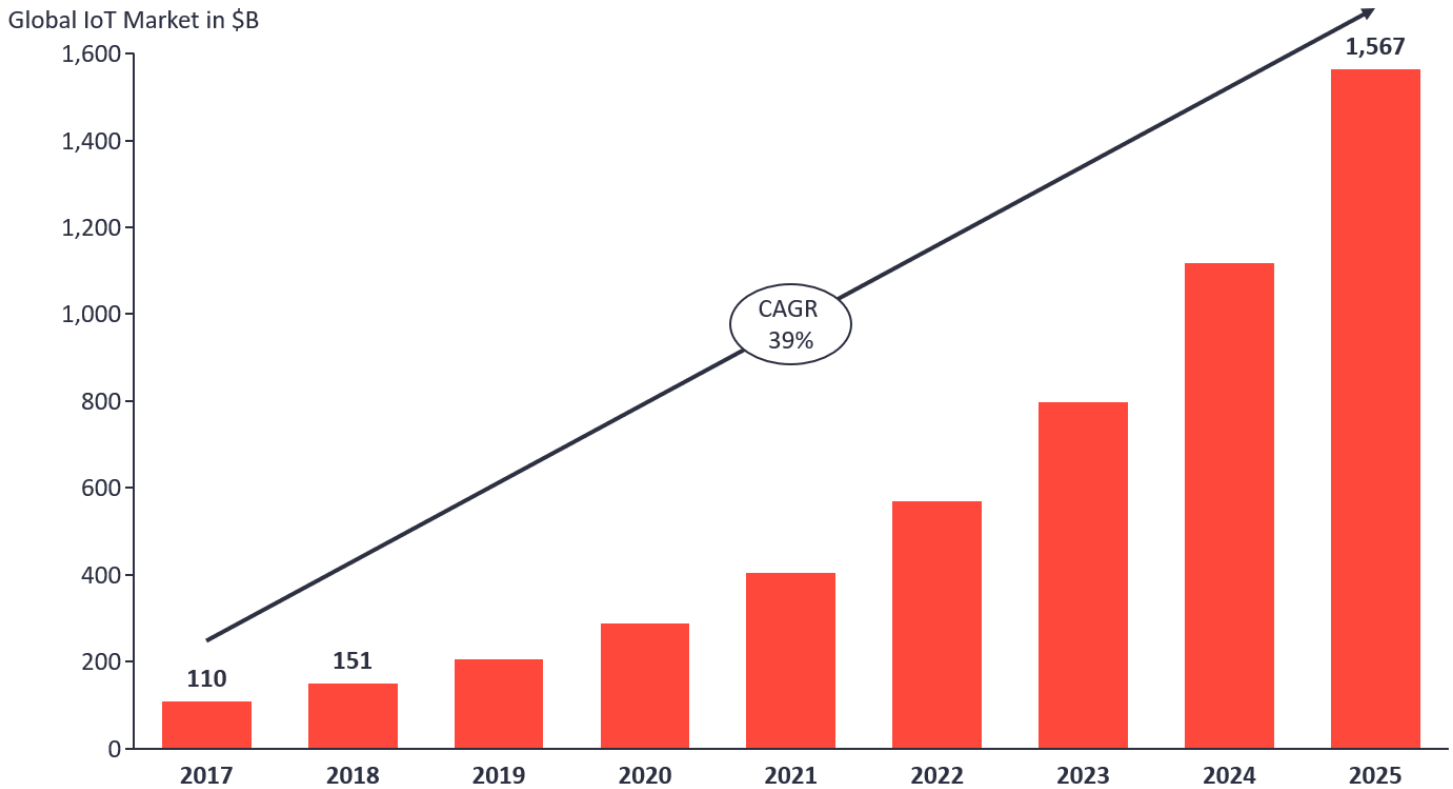
IoT Growth Forecast

IOT ANALYTICS

August 2018

Insights that empower you to understand IoT markets

Global IoT Market Forecast



Note: Market defined as total spend of end-users on IoT solutions
Source: IoT Analytics Research 2018

Job Skills 2020

Top 10 most popular tech skills in 2020*

1. Python
2. React (web)
3. Angular
4. Machine learning
5. Docker
6. Django
7. CompTIA
8. Amazon AWS
9. Deep learning
10. React Native (mobile)

*Ranking is based on highest consumption on Udemy.com in 2019. Source: Udemy Data 2020.

Top 10 emerging tech skills 2020*

1. Gatsby.js (web framework)
2. AWS Big Data
3. React Hooks
4. Microsoft Azure Architecture
5. Next.js (web framework)
6. Apache Airflow (data processing)
7. SwiftUI (Apple apps)
8. Pegasystems (digital process automation)
9. ESP32 (internet of things)
10. Quantum computing

*Data is based on skills with zero consumption in 2018 but surged in 2019 on Udemy. Source: Udemy Data 2020.

Top Tech Skills: Web development, quantum computing, and Internet of Things IoT

A Lot of Jobs - No Skills

- In-Demand Internet of Things (IoT) Skills

"The Internet of things (IoT) is in the midst of an explosion, as more connected devices proliferate. But there's a problem: not enough talent with the right skills to manage and execute on IoT project. So while IoT is poised to be the next great IT jobs boom, insufficient staffing and lack of expertise is the top-cited barrier for organiz currently looking to implement and benefit from IoT, according to research from Gartner."